BayLearn — Question Generation Baseline Analysis

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Purpose: This document establishes the baseline quality of questions generated using the *prompt layer only* approach (no improvement layers). This baseline will be used to measure the impact of adding layers like context enrichment, semantic validation, and quality scoring.

Configuration: All questions generated from uniform study material (Physics fundamentals) using the Groq LLM (llama-3.3-70b-versatile) with temperature=0.85.

EASY DIFFICULTY

Q1: What is the definition of inertia according to Newton's First Law of Motion?

- **A**. The tendency of an object to change its motion
- **B**. The tendency of an object to resist changes in its motion
- **C**. The tendency of an object to move in a circular path
- **D**. The tendency of an object to move in a straight line

Correct Answer: The tendency of an object to resist changes in its motion

Explanation: Newton's First Law of Motion describes the concept of inertia.

Q2: Identify the equation that represents the relationship between work, force, and displacement.

- **A**. $W = F \times d \times \sin(\theta)$
- **B**. $W = F \times d \times \cos(\theta)$
- **C**. $W = F / d \times \cos(\theta)$
- **D**. $W = F + d \times \cos(\theta)$

Correct Answer: $W = F \times d \times \cos(\theta)$

Explanation: The equation for work done is given by the product of force, displacement, and the cosine of the angle between them.

Q3: Define the concept of kinetic energy.

- **A**. The energy of an object due to its position or configuration
- **B**. The energy of an object due to its motion
- **C**. The energy of an object due to its temperature
- **D**. The energy of an object due to its volume

Correct Answer: The energy of an object due to its motion

Explanation: Kinetic energy is the energy associated with the motion of an object.

Q4: List the type of collision where kinetic energy is conserved.

A. Inelastic collision

B. Elastic collision

C. Partial collision

D. Perfect collision

Correct Answer: Elastic collision

Explanation: In an elastic collision, the total kinetic energy is conserved.

Q5: Name the force required to keep an object moving in a circular path.

A. Centrifugal force

B. Centripetal force

C. Gravitational force

D. Frictional force

Correct Answer: Centripetal force

Explanation: The centripetal force is necessary to maintain an object's circular motion.

MEDIUM DIFFICULTY

Q1: Apply Newton's Second Law to a situation where a 50 kg object is accelerating at 2 m/s^2. Describe how to calculate the net force acting on the object.

A. F = (50 kg) * (2 m/s^2) = 100 N

B. F = (50 kg) / (2 m/s^2) = 25 N

C. F = (50 kg) + (2 m/s^2) = 52 N

D. F = (50 kg) - (2 m/s^2) = 48 N

Correct Answer: F = (50 kg) * (2 m/s^2) = 100 N

Explanation: Newton's Second Law states that F = ma, so we multiply the mass (50 kg) by the acceleration (2 m/s^2) to get the net force (100 N).

Q2: Compare and contrast the concepts of displacement, velocity, and acceleration in the context of kinematics. Summarize the key differences between these three quantities.

A. Displacement is a scalar, velocity is a vector, and acceleration is a scalar.

B. Displacement, velocity, and acceleration are all vectors.

C. Displacement is a vector, velocity is a scalar, and acceleration is a vector.

D. Displacement, velocity, and acceleration are all scalars.

Correct Answer: Displacement, velocity, and acceleration are all vectors.

Explanation: In kinematics, displacement, velocity, and acceleration are all vector quantities, meaning they have both magnitude and direction.

Q3: Solve for the work done on an object when a 20 N force acts on it for 5 m in the same direction as the displacement. Explain the formula used to calculate the work done.

A. W = (20 N) * (5 m) * cos(0°) = 100 J

B. W = (20 N) * (5 m) * cos(90°) = 0 J

C. W = (20 N) / (5 m) = 4 J

D. W = (20 N) + (5 m) = 25 J

Correct Answer: W = (20 N) * (5 m) * cos(0°) = 100 J

Explanation: The formula for work is W = F × d × cos(θ), where F is the force, d is the displacement, and θ is the angle between the force and displacement. Since the force and displacement are in the same direction, θ = 0° and cos(0°) = 1.

Q4: Describe the difference between an elastic collision and an inelastic collision in terms of momentum and kinetic energy conservation. Apply this understanding to a scenario where two objects collide.

A. Elastic collisions conserve momentum but not kinetic energy, while inelastic collisions conserve both momentum and kinetic energy.

B. Elastic collisions conserve both momentum and kinetic energy, while inelastic collisions conserve momentum but not kinetic energy.

C. Elastic collisions conserve kinetic energy but not momentum, while inelastic collisions conserve momentum but not kinetic energy.

D. Elastic collisions conserve neither momentum nor kinetic energy, while inelastic collisions conserve both momentum and kinetic energy.

Correct Answer: Elastic collisions conserve both momentum and kinetic energy, while inelastic collisions conserve momentum but not kinetic energy.

Explanation: In an elastic collision, both momentum and kinetic energy are conserved. In an inelastic collision, momentum is conserved, but kinetic energy is not conserved due to energy loss as heat, sound, or deformation.

Q5: Explain the concept of centripetal force and its relationship to gravitational force in the context of orbital motion. Apply this understanding to a satellite orbiting the Earth.

A. Centripetal force is greater than gravitational force in orbital motion.

B. Centripetal force is less than gravitational force in orbital motion.

C. Centripetal force is equal to gravitational force in orbital motion.

D. Centripetal force is not related to gravitational force in orbital motion.

Correct Answer: Centripetal force is equal to gravitational force in orbital motion.

Explanation: In orbital motion, the centripetal force (Fc = mv^2/r) is equal to the gravitational force (F = G(m■m■)/r^2) acting on the satellite, allowing it to maintain a stable orbit around the Earth.

HARD DIFFICULTY

Q1: Analyze the relationship between force, mass, and acceleration in Newton's Second Law of Motion. Evaluate how the mass of an object affects its acceleration when a constant force is applied, and propose a solution to maximize acceleration.

A. Increase the mass to increase acceleration

B. Decrease the mass to increase acceleration

C. Keep the mass constant to maintain acceleration

D. Increase the force to decrease acceleration

Correct Answer: Decrease the mass to increase acceleration

Explanation: Newton's Second Law states that F = ma, so if the force is constant, decreasing the mass will result in increased acceleration.

Q2: Compare the equations of motion for an object under constant acceleration. Justify which equation is most suitable for finding the displacement of an object given its initial velocity, acceleration, and time.

- **A**. $v = u + at$
- **B**. $s = ut + (1/2)at^2$
- **C**. $v^2 = u^2 + 2as$
- **D**. $a = v/t$

Correct Answer: $s = ut + (1/2)at^2$

Explanation: This equation directly relates displacement (s) to the given parameters of initial velocity (u), acceleration (a), and time (t).

Q3: Design an experiment to demonstrate the conservation of energy in a closed system. Evaluate the role of kinetic and potential energy in the experiment and propose a method to measure the total mechanical energy.

- **A**. Measure the change in kinetic energy and ignore potential energy
- **B**. Measure the change in potential energy and ignore kinetic energy
- **C**. Measure both kinetic and potential energy to calculate total mechanical energy
- **D**. Measure the change in thermal energy

Correct Answer: Measure both kinetic and potential energy to calculate total mechanical energy

Explanation: Conservation of energy states that the total mechanical energy (KE + PE) remains constant in a closed system. Measuring both types of energy is necessary to demonstrate this principle.

Q4: Evaluate the differences between elastic and inelastic collisions in terms of momentum and kinetic energy conservation. Propose a solution to determine the type of collision based on the given data.

- **A**. Measure the change in momentum to determine collision type
- **B**. Measure the change in kinetic energy to determine collision type
- **C**. Measure the change in potential energy
- **D**. Measure the change in thermal energy

Correct Answer: Measure the change in kinetic energy to determine collision type

Explanation: In elastic collisions, kinetic energy is conserved, while in inelastic collisions, kinetic energy is not conserved. Measuring the change in kinetic energy can help determine the type of collision.

Q5: Analyze the role of centripetal force in circular motion. Design an experiment to measure the centripetal force acting on an object moving in a circular path and compare it to the gravitational force.

- **A**. Centripetal force is greater than gravitational force
- **B**. Centripetal force is less than gravitational force
- **C**. Centripetal force is equal to gravitational force in orbital motion
- **D**. Centripetal force is not related to gravitational force

Correct Answer: Centripetal force is equal to gravitational force in orbital motion

Explanation: In orbital motion, the centripetal force ($Fc = mv^2/r$) is balanced by the gravitational force ($F = G(m■m■)/r^2$), keeping the object in a stable circular path.

Analysis Notes

Key Observations:

1. **Prompt Layer Only:** These questions represent the baseline output of the prompt engineering layer without any additional refinement layers (context enrichment, semantic validation, quality scoring).

2. **Expected Improvement Areas:**
• Context Enrichment Layer: Could improve question relevance by selecting better study material chunks
• Semantic Validation Layer: Could ensure questions are factually correct and well-formed
• Quality Scoring Layer: Could filter out low-quality questions before returning to user

3. **Consistency Metrics:**
• Question Variety: Assess if questions cover different concepts
• Difficulty Alignment: Verify that easy/medium/hard questions match their intended level
• Answer Clarity: Check if correct answers are unambiguous

4. **Next Steps:**
• Implement context enrichment layer to improve material relevance
• Add semantic validation to verify question quality
• Generate new baseline after implementing each layer for comparison